Decidable Type-Checking for the Dependent Categorical Combinators

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Abstract

We present an extension of the categorical combinators [4] to symmetric monoidal categories, enabling affine dependent types of the nonlinear categorical combinator terms. We formalize the metatheory in the Lean 4 proof assistant. We prove several important properties of the calculus, including that type-checking is decidable, and that the affine restriction of the combinators is strongly-normalizing, and globally confluent. We achieve judgmental decidability in principle by showing that the types of all objects can be expressed using the affine restriction, which we demonstrate is strongly-normalizing by a syntactic decreasing measure. We outline key challenges and architectural strategies for formalizing a dependent combinator calculus with such properties in a proof assistant, and describe our progress towards compiling a λ-calculus with dependent types (Π-types and Σ-types) to our combinators.

1 Introduction

Simply-typed combinatory logics like the SK-combinators are known to be equivalent to the simply-typed λ-calculus [1]. These calculi can avoid contexts, and might enjoy benefits as such (e.g., unification). Work has been done on extending these logics with dependent types [2], but no implementation has been proposed, nor have such theories been formalized in a proof assistant.

Combinatory logics suffer from the known disadvantage that they are an inefficient compilation target. For example, when compiling the λ-calculus to the SK-combinators, terms explode in size due to lacking variable substitution [6, p. 267].

Numerous alternative combinatory compilation targets have been proposed. One such calculus is the categorical combinator calculus [4]. The categorical combinators are best described as a literal interpretation of cartesian closed categories as a Lisp-like Debruijn-indexed machine (the Categorical Abstract Machine [3]). Variable substitution is modeled intuitively as composing binary product projections [3, p. 6] over an environment of nested tuples. The categorical combinators have been extended with simple types [4]. There has also been some investigation into extending them to a dependent type theory similar to the Calculus of Constructions [7], but no implementation has been defined.

In this paper, we formalize a metatheory of a dependently-typed categorical combinator calculus with affine types and non-linear terms based on symmetric monoidal categories. We show that type inference is decidable and deterministic. We accomplish this goal in principle by observing that the types of the dependent non-linear categorical combinators are encodable with an affine subset of the usual reduction rules. By showing that the affine restriction of the calculus is strongly normalizing through a syntactic decreasing measure, we achieve total affine β-reduction. We show that the calculus satisfies global confluence by Newman's lemma [5] using our measure. We prove completeness and soundness of affine β-reduction, and soundness of β-normal forms with respect to our reduction relation, and hence achieve decidable affine equality. We are able to avoid a conversion rule in the typing judgment by eagerly normalizing types, and thus achieve deterministic, decidable type inference. We do not assume Π-extensionality, but describe η-reduction rules used for a weak form of extensionality as in Curien et al. [4, p. 221]. We outline several of our architectural decisions including proof automation with typeclasses critical to satisfying our formalization goals in Lean 4.

2 Background

2.1 The Untyped Categorical Combinators

The categorical combinators [4] are the basis of the Categorical Abstract Machine [3]. They are a potential compilation target for the λ-calculus [3]. Categorical combinators simulate variable substitution through projections over nested tuples ("couples", or binary products) constituting an "environment". Binders are simulated through a curry combinator (Λ) which applies its two arguments as a couple to the body ($\Lambda(x)$). The reduction rules for the calculus resemble the rules of the SK combinator calculus, and their categorical semantics are given by cartesian-closed categories [3, p. 178].

(ass)	$(x \circ y)z = x(yz)$
(fst)	$Fst(a, b) = a$
(snd)	$Snd(a, b) = b$
(dpair)	$\langle x, y\rangle z = (xz, yz)$
(curr)	$\Lambda(x)yz = x(y, z)$
(app)	$App(f, x) = fx$
(quote)	$`(x)y = x$

Figure 1: Reduction rules of the untyped combinators as seen in the Categorical Abstract Machine.

2.2 The Simply-Typed Categorical Combinators

Curien extends the combinators with type annotations [4, p. 220], enabling simple typing. Quotation is omitted in the simply-typed variant. The inference rules are as such:

$$\textsc{Id}\ \frac{\alpha : \text{Type}}{Id(\alpha) : \alpha \to \alpha} \qquad \textsc{Coup}\ \frac{x : \alpha, y : \beta}{(x, y) : \alpha \times \beta} \qquad \textsc{Comp}\ \frac{x : \beta \to \gamma, y : \alpha \to \beta}{(x \circ y) : \alpha \to \gamma}$$

$$\textsc{Fst}\ \frac{\alpha : \text{Type}, \beta : \text{Type}}{Fst(\alpha, \beta) : \alpha \times \beta \to \alpha} \qquad \textsc{Snd}\ \frac{\alpha : \text{Type}, \beta : \text{Type}}{Snd(\alpha, \beta) : \alpha \times \beta \to \beta} \qquad \textsc{Pair}\ \frac{x : \alpha \to \tau_1, y : \alpha \to \tau_2}{\langle x, y\rangle : \alpha \to \tau_1 \times \tau_2}$$

$$\textsc{Curr}\ \frac{x : (\alpha \times \beta) \to \tau}{\Lambda(x) : \alpha \to \beta \to \tau} \qquad \textsc{Appl}\ \frac{\alpha : \text{Type}, \beta : \text{Type}}{App(\alpha, \beta) : ((\alpha \to \beta) \times \alpha) \to \beta} \qquad \textsc{App}\ \frac{f : \alpha \to \beta x : \alpha}{fx : \beta}$$

Figure 2: Inference rules of the simply-typed categorical combinators

Curien's categorical combinators are nominally combinatory, but differ from SK and $BCKI$ calculi in that points are still present. Type arguments are given as points, projection mappings in pairs are given as points ($\langle f, g\rangle$), the Curry combinator is defined by a point ($\Lambda(x)$), couple members are specified as points ((a, b)), and composition is pointed with respect to f and g ($(f \circ g)x$). This design is quite natural when interpreting projection as corresponding to Debruijn indices. Take an example reduction in the untyped calculus:

$$\begin{aligned}
&\Lambda(\Lambda(App \circ \langle Snd \circ Fst, App \circ \langle App \circ Fst, Snd\rangle\rangle))nfx \\
&= \Lambda(App \circ \langle Snd \circ Fst, App \circ \langle App \circ Fst, Snd\rangle\rangle)(n, f)x \\
&= (App \circ \langle Snd \circ Fst, App \circ \langle App \circ Fst, Snd\rangle\rangle)((n, f), x) \\
&= App(f, App(App(n, f), x)) \\
&= f(nfx)
\end{aligned} \tag{1}$$

Despite every visible operation being bound by a point, arguments in scope can be rearranged through projections, composition, pairing, and application (evaluation in the CCC).

3 Dependent Categorical Combinators

Our primary insight when extending the categorical combinators to dependent types is that the types of all of the categorical combinators are representable without making use of the pair rule. That is, their types are affine. We refer to the subset of the categorical combinators excluding the pair rule as the "affine restriction." By showing that the affine restriction is strongly normalizing, we gain decidable judgmental equality ($\equiv$) and a decision procedure for type-checking.

We interpret Σ-types as a telescope context as usual. As in the Curry combinator, we specify combinator type arguments as points that internalize an indexing of the environment. For example, the Π combinator is inferred like such:

$$\frac{\textsc{Pi}}{\delta : \text{Sort } m, f_\alpha : \alpha \to \text{Sort } n, f_\beta : \Pi(s : \delta)(x : f_\alpha s), \text{Sort } o}{\Pi(f_\alpha, f_\beta) : \delta \to \text{Sort } o} \tag{2}$$

We interpret couples as dependent sums. Σ-couples make traversing data in a well-typed way quite convenient. All of our objects are well-typed, and Π elimination sorts can be computed with no extra assumptions. We define most of the type arguments of the combinators as functions of the environment eliminating into Sort n. By specifying type arguments in this way, some potential duplication of items in the context can be avoided (e.g., in Π, where the environment is shared across f_α and f_β), thus further enabling a purely affine type theory.

3.1 Σ-Couples

Recall the usual typing of a Σ-pair:

$$\frac{\Gamma \vdash \alpha : \text{Sort } m, \beta : \alpha \to \text{Sort } n}{\Gamma \vdash \Sigma(\alpha, \beta) : \text{Sort } max(m, n)} \tag{3}$$

A Σ-type can refer to variables in its fst and snd types, and fst itself in the type of snd. Σ-types are one of the canonical dependent types, alongside dependent products. There is a known equivalence between Π and Σ in that a function $\Pi(x : \alpha)(y : \beta x).\gamma x y$ can be transformed into a dependent function $\Pi(s : \Sigma(x : \alpha), \beta x), \gamma fst(s) snd(s)$, and vice-versa ("currying"-"uncurrying").

In the untyped categorical combinators, Uncurry can be written as:

$$\begin{aligned} Curry(x) &:= \Lambda(x) \\ Uncurry(x) &:= App \circ \langle x \circ Fst, Snd \rangle \\ Uncurry(x)(y, z) &= xyz \end{aligned} \tag{4}$$

Naively interpreting couples as Σ pairs with point-free type arguments makes this example (**??**) quite difficult, especially in an affine setting. In $Uncurry(x)$, $snd\ (z : \beta y)$ depends on the fst component $(y : \alpha)$ of the encapsulating couple's preimage, despite y being destroyed upon reduction. App has no knowledge of y, and thus cannot compute the type of (xy, z). We remedy this by indexing Σ with functions of the environment $f_\beta : \Pi(s : \delta)(x : f_\alpha s), \text{Sort } n$ (5), similarly to in Curry ($\Lambda(x)$).

$$\frac{\textsc{Sigma}}{\alpha : \text{Sort } m, f_\beta : \delta \to \alpha \to \text{Sort } n}{\Sigma(f_\beta) : \delta \to \text{Sort } max(m, n)} \tag{5}$$

Figure 3: Where δ denotes an arbitrary environment and α denotes the fst component. $\Sigma(f_\beta)(env : \delta) : \text{Sort } max(m, n)$

We thus define couples $[\![\beta]\!](a, b)$ as a degenerate case of Σ where β depends only on fst and not the environment. We represent a no-op indexing of the environment with a unit type $\top$ so as to avoid universe level arguments to Σ. We use the notation $Coup(\beta)$ to refer to the type of a couple where $\beta : \alpha \to \text{Sort } n$ (**??**). We always refer to couples with their β points, since they cannot be inferred.

$$\frac{}{\top : \text{Sort } 0}\ \text{Top} \qquad \frac{}{* : \top}\ \text{Unit} \qquad \frac{A : \text{Sort } m, B : A \to \text{Sort } n, x : A, y : B(x),}{[\![B]\!](x,y) : \Sigma(\text{`(Sort } m, B))A}\ \text{Coup} \qquad \frac{}{Coup(B) \coloneqq (\Sigma(\text{`}(\top, B))*)}$$

This view of couples as Σ-types naturally encodes a telescope context, and proves to be invaluable when compiling dependently-typed λ-calculi to the categorical combinators. A context Γ can be encoded as a nested couple:

$$x_0 : A_0, x_1 : A_1(x_0), x_2 : A_2(x_0, x_1) \approx [\![\Sigma(A_1), A_2]\!]([\![A_1]\!](x_0, x_1), x_2) \tag{6}$$

We make use of this isomorphism to represent contexts in our ongoing effort to compile dependent λ-calculus to our combinators.

3.2 Type Arguments Index the Environment

All combinator objects have type annotations specified as points (usually as families indexing the environment). For example, the projection $Snd(f_\beta)$ is indexed by the term f_β which depends on an arbitrary environment $(s : \delta)$ and the term $fst : \alpha$. Couple projection can be derived, and we refer to it as $Sndc(\beta)$.

3.3 Inference Rules

We make use of an infinite sort hierarchy to avoid paradoxes (i.e., $\text{Sort } m : \text{Sort } m + 1$). Arrows are not first-class, and can be defined with Π. We do not explicitly apply the environment to the codomain in Π in our written inference rules. The full types in our deep embedding are accessible in the appendix (A).

3.3.1 Application and Π

The inference rule for function application is as usual, except the codomain β is applied the environment $s : \delta$, and the argument $x : f_\alpha s$. Since $f_\alpha s$ is discarded after type-checking, the environment can safely traverse the Π object affinely (see inference rule (2)). We normalize types eagerly, but we do not always indicate so in our inference rules (see (A)). We use $\rightarrowtail^!$ to denote star-step affine reduction to a normal form.

$$\frac{f : \Pi(f_\alpha, f_\beta)s, f_\alpha x \rightarrowtail^! \alpha \qquad x : \alpha_2, \alpha_2 \rightarrowtail^! \alpha \qquad f_\beta s x \rightarrowtail^! \beta}{fx : \beta}\ \text{App} \tag{7}$$

Figure 4: Inference rule: application

Since we do not make use of variables, Π is itself a combinator dependent on an environment. Since the dependency is internalized by f_α and f_β, its type can be expressed with an arrow (2).

3.3.2 Arrow types

We define arrows $(A \to B)$ as Π objects quoting their codomain to discard the environment-argument couple (8).

$$\frac{A : \text{Sort } m}{A \to B \coloneqq (\Pi(Id(\text{Sort } m), \Lambda(\text{Sort } m, Id(\text{Sort } m), \text{`}(Coup(Id(\text{Sort } m)), B)))A)} \tag{8}$$

3.3.3 Couple Projections

We define dependent and nondependent versions of the *Snd* projection to bootstrap several of the combinators' types. We also derive a *Sndc* combinator which accepts a $Coup(\beta)$ not depending on an environment. $Fst(\beta)$ depends on an environment, and *Fstc* is similarly derived. Composition $(f \circ g)$

is nondependent with respect to g, although f may depend on gx. A form of semi-dependency can be derived if g carries its dependency in a couple $(gx : \Sigma(\ldots)\ldots)$.

We use thunking to reduce pairs ($\langle f, g\rangle x$) lazily. That is, $\langle f, g\rangle x$ is a normal form, and projections and evaluation are defined on it as if it were the reduced couple (fx, gx). This approach enables some usage of the pair rule in our types, though we do not use it to define any of our core types.

The inference rules for projection on Σ-terms (couples and pairs) are as such:

$$\textsc{Fst}\ \frac{\alpha : \text{Sort } m, \delta : \text{Sort } o, f_\beta : \delta \to \alpha \to \text{Sort } n}{Fst(\beta) : \Pi(env : \delta)(s : \Sigma(f_\beta)env), \alpha} \qquad \textsc{Snd}\ \frac{\alpha : \text{Sort } m, \delta : \text{Sort } o, f_\beta : \delta \to \alpha \to \text{Sort } n}{Snd(\beta) : \Pi(env : \delta)(s : \Sigma(f_\beta)env), f_\beta env(Fst(f_\beta)envs))}$$

$$\textsc{Snd'}\ \frac{\alpha : \text{Sort } m, \beta : \text{Sort } n}{Snd'(\alpha, \beta) : Coup(`(\alpha, \beta)) \to \beta} \qquad \frac{}{Sndc(\beta) \coloneqq Snd(`(\top, \beta))*} \qquad \frac{}{Fstc(\beta) \coloneqq Fst(`(\top, \beta))*}$$

Our inference rules for projection apply to applied pairs as well, since they conceptually reduce to couples.

3.3.4 Identity

Id accepts α as a point, but not a function of the environment. A type-point-free variant of Id can be derived easily with Curry (Λ). The type of $Id(\alpha)$ is an arrow (9).

$$\textsc{Id}\ \frac{\alpha : \text{Sort } m}{Id(\alpha) : \alpha \to \alpha} \tag{9}$$

3.3.5 Composition

We have nondependent composition as in Curien's combinators, except that the codomain of f can depend on gx, but the codomain of g cannot depend on x (10). Allowing both f and g to depend on their inputs would require duplicating the argument x in our reduction rules, whereas x is not usually duplicated in the categorical combinators.

$$\textsc{Comp.}\ \frac{f : \Pi(x : \beta), \gamma x, g : \beta \to \beta}{(f \circ g) : \Pi(x : \alpha), \gamma(gx)} \tag{10}$$

Figure 5: Inference rule: composition

Full dependency can be recovered when g produces a Σ-term encoding its dependency (e.g., $g : \alpha \to Coup(\beta)$).

3.3.6 Pairs

Recall $\langle f, g\rangle x = (fx, gx)$. We do not define reduction directly on pairs, and instead give projections and evaluation rules for $\langle f, g\rangle x$ as if it were the reduced couple.

$$\textsc{Pair}\ \frac{f : \Pi(x : \alpha), \beta x \qquad g : \Pi(x : \alpha), \gamma x}{\langle f, g\rangle : \Pi(x : \alpha), \Sigma(\Lambda(\alpha, \beta, \gamma\ Fstc(\beta)))x} \tag{11}$$

3.3.7 Quotation

The quotation combinator ($`(\beta, x)$) is not dependent, but a dependent version similar to dependent K can be derived.

$$\textsc{Quot}\quad \frac{\beta : \text{Sort } n, x : \alpha}{`(\beta, x) : \beta \to \alpha} \tag{12}$$

3.3.8 Curry

Recall $\Lambda(x)yz = x(y, z)$. Our inference rule for application (17) proves to be useful in defining the type, since terms in context can be used more than once in a Π tree, so long as they are used at most once in each argument type.

$$\textsc{Curry}\quad \frac{\alpha : \text{Sort } m, \beta : \alpha \to \text{Sort } n, x : (\Pi(s : Coup(\beta)), \gamma s)}{\Lambda(\alpha, \beta, x) : \Pi(x : \alpha)(y : \beta x), \Lambda(\alpha, \beta, \gamma)xy} \tag{13}$$

Figure 6: Inference rule: curry

3.3.9 App and Subst Combinators

As in Curien's combinators, the *App* combinator evaluates a couple as a computation. Its reduction rule is:

$$App(f, x) \rightarrowtail fx \tag{14}$$

Defining the type of *App* proves to be quite difficult in this configuration, since $[\![\beta]\!](f, x)$ is a Σ-couple. The term x does not actually depend on f. The book-keeping required to discard f in β can be easily avoided by adopting the rule $App(x, f) \rightarrowtail (fx)$, as in Seely [8, p. 39].

We mirror the inference rule for Π in that the types points for *App* index an environment. In fact, the types of the points f_α and f_β are the exact same as in Π. Since *App* indexes the environment, we define its reduction rule as:

$$App(f_\alpha, f_\beta)env[\![\ldots]\!](x, f) \rightarrowtail fx \tag{15}$$

The inference rule requires duplicating the environment in the codomain of $App(\ldots, \ldots)env[\![\ldots]\!](x, f)$, since f_β depends on the environment and the Fst projection x, which also depends on the environment. However, the environment is not computationally relevant to the result of projection, and behaves more like a context. That is, duplicating the environment corresponds directly to context substitution. We can avoid the duplication altogether by defining *Subst* with respect to the combinator depending on the environment. The reduction and inference rule is defined for Fst like such:

$$\begin{array}{c} Subst(x, Fst(f_\beta))env[\![\ldots]\!](y, z) \rightarrowtail xenvy \\ Subst(x, Fst(f_\beta))env[\![\ldots(\langle f, g\rangle z) \rightarrowtail xenv(fz) \end{array} \qquad \textsc{Subst Fst}\quad \frac{x : \Pi(env : \delta)(x : \alpha), \gamma envx}{Subst(x, Fst(f_\beta)) : \Pi(env : \delta)(s : \Sigma(f_\beta)), Subst(\gamma, Fst(f_|beta))} \tag{16}$$

We define the *App* inference rule (17) using our *Subst* combinator.

$$\textsc{Appl}\quad \frac{\delta : \text{Sort } m, f_\alpha : \delta \to \text{Sort } m, f_\beta : \Pi(s : \delta)(x : f_\alpha s), \text{Sort } o}{t_{\beta xy} \coloneqq \Lambda(\delta, f_\alpha, \Pi(f_\alpha, f_\beta) \circ Fstc(f_\alpha)), App(f_\alpha, f_\beta) : \Pi(env : \delta)(s : \Sigma(t_{\beta xy})env), Subst(f_\beta, Fst(t_{\beta xy}))} \tag{17}$$

Figure 7: *App* inference rule

4 Affine and η-Reduction

In general, dependent combinatory logics require complex types. Hence, we extend Curien's η-reduction rules for the categorical combinators to enable some examples [4, p. 221]. Since our types are affine (3.3), we can realize decidable type-checking by demonstrating strong normalization of $\eta^!$ and $\rightarrowtail^!$ and expressing conversion in our judgment relation as composition of $\rightarrowtail^!$ and $\eta^!$.

We prove soundness and completeness of reduction functions for each scheme. We demonstrate strong normalization by a syntactic decreasing measure.

4.1 Automating Completeness

The derivations of completeness and soundness for $\rightarrowtail$ and η-reduction are structurally quite similar. Thus, we make the practical design decision to abstract the notions of a size and step-like relations with Lean's typeclass system.

We identify a set of minimal requirements to prove completeness and strong normalization for our rules: soundness of a computable definition with respect to the inductive relation, a proof that a size measure decreases over the reduction, local confluence of the respective single-step relation, and "semi-completeness". We define these properties as typeclasses and instantiate them for η and $\rightarrowtail$. In doing so, we automatically inherit global confluence by Nemann's lemma, a total, terminating reduction function, strong normalization, well-foundedness with respect to the relation, uniqueness of normal forms, completeness of the reduction function with respect to $\rightarrow^!$, and decidability of the reduction relation up to normal forms.

Our notion of a relation requires an inductive relation and an implementation with soundness:

```
class Rel.{u, v, s} {α : Type u} {β : Type s} (r : α → β → Type v)
  (m : Type (max v s) → Type (max v s) := Id) [hM : Monad m] where
  run (x : α) : m ((y : β) × (r x y))
  sound       : run x = pure ⟨y, h⟩ → r x y := fun _ => h
```

We use universe and monadic polymorphism to express single-step reduction with `Rel`, since single-step reduction is not always defined.

We define completeness with respect to a relation (`Rel`) as such:

```
open Rel in class Complete.{u, v, s} {α : Type u} {β : Type s} (r : α → β → Sort v)
  (P : β → Prop := fun _ => True)
  (m : Type (max v s) → Type (max v s) := Id) [hm : Monad m]
  [hR : Rel.{u, v, s} (fun (x : α) (y : β) => PULift.{v + 1, v} (r x y)) (m := m)] where
  complete : ∀ (h : r x y) (_p : P y), hR.run x = hm.pure ⟨y, PULift.up h⟩
```

We define a the general form of a size measure, automatically derive completeness with tactics:

```
open Rel in class Measure (r : Expr → ℕ → Prop)
  extends Rel.{0, 0, 0} (fun x n => PULift.{1, 0} (r x n)), Complete.{0} r where
  complete := by
    intro x y h _h
    induction h
    all_goals (
      simp_all [pure, id_pure_ext]
      try rfl
      try grind
    )

abbrev Measure.size (t : outParam (Expr → ℕ → Prop)) [hM : Measure t]
  (e : Expr) : ℕ := (hM.run e).fst
```

We define a reduction relation as a relation satisfying the decreasing measure property. We can usually derive the proof using tactics.

```
class RedRel (r : Expr → Expr → Prop) (t : outParam (Expr → ℕ → Prop))
  (m : Type → Type := Option) [_hM : Monad m]
  [hM : Measure t] extends Rel.{0, 0, 0} (fun (x : Expr)
  (y : Expr) => PULift.{1, 0} (r x y)) (m := m) where
  decreasing : r x y → (hM.run y).fst < (hM.run x).fst := by
    intro x y h
    induction h
    all_goals (
      try dsimp
```

```
      try simp_all [dcc_rel, dcc_size]
      try grind
      try aesop (rule_sets := [DccRel, DccSize])
      try rfl)
```

Relations with a decreasing measure inherit strong normalization, well-foundedness, and a total reduction function:

```
def RedRel.sn {r : Expr → Expr → Prop} (t : outParam (Expr → ℕ → Prop))
  {m : Type → Type} [Monad m] [hM : Measure t]
  (hR : RedRel r t (m := m) (hM := hM)) (e : Expr) : Acc (fun b a => r a b) e :=
    Acc.intro e (fun e' hr => by
    have h := hR.decreasing hr
    exact sn t hR e')
termination_by (hM.run e).fst

def RedRel.wellFounded {r : Expr → Expr → Prop} (t : outParam (Expr → ℕ → Prop))
  {m : Type → Type} [Monad m] [hM : Measure t]
  (hR : RedRel r t (m := m) (hM := hM)) : WellFounded (fun a b => r b a) := ⟨hR.sn⟩

def reduce {t : Expr → ℕ → Prop} [hT : Measure t] (r : Expr → Expr → Prop)
  [hR : RedRel r t (hM := hT)] (e : Expr) :
  ((e' : Expr) × (PULift.{1, 0} (Relation.ReflTransGen r e e'))) :=
  match hR.run e with
  | .some ⟨e', ⟨h⟩⟩ =>
    have _size_ : (hT.run e').fst < (hT.run e).fst := hR.decreasing h
    let ⟨e'', ⟨h₂⟩⟩ := reduce r (t := t) e'
    ⟨e'', ⟨.trans (.single h) h₂⟩⟩
  | .none => ⟨e, ⟨.refl⟩⟩
termination_by (hT.run e).fst

@[dcc_reduce, inline] abbrev reduce' {t : Expr → ℕ → Prop} [hT : Measure t]
  (r : Expr → Expr → Prop) [hR : RedRel r t (hM := hT)]
  (e : Expr) : Expr := (reduce r e).fst
```

Since $\rightarrowtail$ has congruence, confluence is required to show completeness of `reduce`. We define the notion like such:

```
open Rel in class Confluent.{u} {α : Type u} (r : α → α → Prop)
  (m : (α → α → Prop) → α → α → Prop := fun x => x) where
  conf : (m r A B) → (m r A C) → ∃ D, Relation.ReflTransGen.{u} r B D
  ∧ Relation.ReflTransGen.{u} r C D
```

By Newmann's lemma, we derive global confluence and uniqueness of normal forms for relations with a measure.

Theorem 1. *If r is locally confluent and strongly normalizing by a decreasing measure, then r is globally confluent.*

Proof. By Newmann's lemma:

```
open Confluent in def RedRel.globally_confluent
  {r : Expr → Expr → Prop} {t : Expr → ℕ → Prop} {m : Type → Type}
  [hM : Measure t] [Monad m] [hC : Confluent r]
  [hR : RedRel r t (m := m) (hM := hM)] : (Confluent r (m := Relation.ReflTransGen)) where
  conf := fun {A B C} (h₁ : hR.ReflTrans A B) (h₂ : hR.ReflTrans A C) => by
    induction A using hR.wellFounded.induction generalizing B C with
    | h x ih =>
      match h₁.cases_head, h₂.cases_head with
      | .inl heq₁, .inl heq₂ =>
        exact ⟨x, heq₁ ▸ .refl, heq₂ ▸ .refl⟩
      | .inl heq₁, _ =>
        exact ⟨C, heq₁ ▸ h₂, .refl⟩
      | _, .inl heq₁ =>
        exact ⟨B, .refl, heq₁ ▸ h₁⟩
      | .inr ⟨C₁, hr₁, hrt₁⟩, .inr ⟨C₂, hr₂, hrt₂⟩ =>
        have ⟨D, hrt₁b, hrt₂b⟩ := hC.conf hr₁ hr₂
        have ⟨_, hrtt₁, hrtt₁b⟩ := ih _ hr₁ hrt₁ hrt₁b
        have ⟨_, hrtt₂, hrtt₂b⟩ := ih _ hr₂ hrt₂ hrt₂b
        have ⟨_, h₁, h₂⟩ := ih _ hr₁ (hrt₁b.trans hrtt₁b) (hrt₁b.trans hrtt₂b)
        exact ⟨_, .trans hrtt₁ h₁, .trans hrtt₂ h₂⟩
```

□

Theorem 2. *If r confluent, and rAC and rAB and B and C are normal forms, then $B = C$.*

Proof.

```
def IsNorm.unique_refl {r : Expr → Expr → Prop} (hn : IsNorm r A)
  (hr : (Relation.ReflTransGen r) A B) : A = B := by
  rw [← Relation.reflGen_transGen] at hr
  cases hr with
  | refl => rfl
  | single h => exact hn.unique_trans h

open Confluent in @[dcc_rel, aesop safe destruct (rule_sets := [DccRel])]
  theorem norm_unique {r : Expr → Expr → Prop} {m : (Expr → Expr → Prop) → Expr → Expr → Prop}
  [hC : Confluent r m] (h₁ : m r A B) (hn₁ : IsNorm r B) (h₂ : m r A C) (hn₂ : IsNorm r C) : B
  ↪  = C := by
  have ⟨D, hr₁, hr₂⟩ := hC.conf h₁ h₂
  have ha := hn₁.unique_refl hr₁
  have hb := hn₂.unique_refl hr₂
  simp_all
```

□

We require a weak form of completeness to derive full completeness for `reduce`: if $A \to B$, then there exists some C and a proof h produced by `run` that $A \to C$:

```
open Rel in class SemiComplete {t : Expr → ℕ → Prop} [hM : Measure t]
  (r : Expr → Expr → Prop)
  [hR : RedRel r t (m := Option) (hM := hM)] where
  complete : r A B → ∃ B h, (hR.run A) = .some ⟨B, ⟨h⟩⟩
```

Relations with a measure, semi-completeness, and confluence inherit completeness of `reduce` with respect to normal forms.

Theorem 3. *Suppose $\to_r$ is strongly normalizing under a decreasing measure, semi-complete, and confluent. Then, if hr is a proof that $A \to_r^* B$ and B is a normal form, then `reduce A = (B, hr)`.*

Proof.

```
theorem reduce_complete {r : Expr → Expr → Prop} {t : Expr → ℕ → Prop}
  [hM : Measure t] [hR : RedRel r t (m := Option.{0}) (hM := hM)] [hN : SemiComplete r (hR :=
  ↪  hR)] [Confluent r]
  {A B : Expr} (hr : Relation.ReflTransGen r A B) (hn : IsNorm r B) : (reduce r A) = ⟨B, ⟨hr⟩⟩
  ↪  := by
  simp [reduce]
  split
  case h_1 B' hr₂ heq =>
    have ⟨C, h₁, h₂⟩ := hR.globally_confluent.conf hr (.single hr₂)
    have h_b_c_eq := hn.unique_refl h₁
    have hr_b'_b := h_b_c_eq ▸ h₂
    match hr_b'_b.cases_head with
    | .inl h =>
      have h_red_norm := hR.reduce_norm_fst hn
      cases h_b_c_eq; cases h
      congr
      repeat simp_all
    | .inr ⟨_, hrB', hrt⟩ =>
      rename_i w
      have h_dec : (hM.size t B') < (hM.size t A) := by
        have h₁ := hR.decreasing hr₂
        have h₂ := hR.decreasing hrB'
        grind
      have h := reduce_complete (.trans (.single hrB') hrt) hn
      have h_fst : (reduce r B').fst = B := by
        simp_all
      cases h_fst
      congr
  case h_2 hneq =>
    match hr.cases_head with
    | .inl h => cases h; congr
    | .inr ⟨_, hrb', hrt⟩ =>
      have ⟨_, hA₁, hA₂⟩ := hN.complete hrb'
      simp_all
termination_by ((hM.size t A), (hM.size t B))
```

□

4.2 Affine Reduction Rules

Our affine reduction rules are mostly the same as in [3], except we omit the general pair rule, define a context substitution combinator *Subst* (16), and flip the order of the couple argument to *App*:

```
inductive RedAff : Expr → Expr → Prop
  | subst₁p  : RedAff (DCC (Subst(x, Fst(f_β)) env (⟨f, g⟩ z))) (DCC (x env (f z)))
  | subst₁   : RedAff (DCC (Subst(x, Fst(f_β)) env ⟦β⟧(y, z))) (DCC (x env y))
  | subst₂   : RedAff (DCC (Subst(x, Snd(f_β)) env ⟦β⟧(y, z))) (DCC (x env z))
  | snd'     : RedAff (DCC (Snd'(_δ, _τ) ⟦β⟧(a, b))) b
  | quot     : RedAff (DCC (`(β, x) y)) x
  | id       : RedAff (DCC (Id(α) x)) x
  | appl     : RedAff (DCC (App(_α, _β) env ⟦β⟧(x, y))) (DCC (x y))
  | fst      : RedAff (DCC (Fst(_α) _env ⟦β⟧(a, b))) a
  | fstp     : RedAff (DCC (Fst(_α) _env (⟨f, g⟩ x))) (DCC (f x))
  | snd      : RedAff (DCC (Snd(_α) _env ⟦β⟧(a, b))) b
  | sndp     : RedAff (DCC (Snd(_α) _env (⟨f, g⟩ x))) (DCC (g x))
  | comp     : RedAff (DCC ((x ∘ y) z)) (DCC (x (y z)))
  | curr     : RedAff (DCC (Λ(α, β, x) y z)) (DCC (x ⟦β⟧(y, z)))
  | sigl     : RedAff x x' → RedAff (DCC ⟦β⟧(x, y)) (DCC ⟦β⟧(x', y))
  | sigr     : RedAff y y' → RedAff (DCC ⟦β⟧(x, y)) (DCC ⟦β⟧(x, y'))
  | congrFun : RedAff f f' → RedAff (DCC (f x)) (DCC (f' x))
  | congrArg : RedAff x x' → RedAff (DCC (f x)) (DCC (f x'))
```

We compute single-step affine reduction like such:

```
def step? (e : Expr) : Option ((e' : Expr) × (PULift.{1, 0} (RedAff e e'))) :=
  match e with
  | (DCC (Subst(_x, Fst(_f_β)) _env (⟨_f, _g⟩ _z))) => .some ⟨_, ⟨.subst₁p⟩⟩
  | (DCC (Subst(_x, Fst(_f_β)) _env ⟦_β⟧(_y, _z))) => .some ⟨_, ⟨.subst₁⟩⟩
  | (DCC (Subst(_x, Snd(_f_β)) _env ⟦_β⟧(_y, _z))) => .some ⟨_, ⟨.subst₂⟩⟩
  | (DCC (Snd'(_α, _τ) ⟦_β⟧(_a, _b)))              => .some ⟨_, ⟨.snd'⟩⟩
  | (DCC (`(_β, _x) _y))                           => .some ⟨_, ⟨.quot⟩⟩
  | (DCC (Id(_α) _x))                              => .some ⟨_, ⟨.id⟩⟩
  | (DCC (App(_α, _β) _env ⟦_β₂⟧(_x, _y)))         => .some ⟨_, ⟨.appl⟩⟩
  | (DCC (Fst(_α) _env ⟦_γ⟧(_a, _b)))              => .some ⟨_, ⟨.fst⟩⟩
  | (DCC (Fst(_α) _env (⟨_f, _g⟩ _x)))             => .some ⟨_, ⟨.fstp⟩⟩
  | (DCC (Snd(_α) _env ⟦_γ⟧(_a, _b)))              => .some ⟨_, ⟨.snd⟩⟩
  | (DCC (Snd(_α) _env (⟨_f, _g⟩ _x)))             => .some ⟨_, ⟨.sndp⟩⟩
  | (DCC ((x ∘ y) z))                              => .some ⟨(DCC (x (y z))), ⟨.comp⟩⟩
  | (DCC (Λ(_α, β, x) y z))                        => .some ⟨(DCC (x ⟦β⟧(y, z))), ⟨.curr⟩⟩
  | DCC ⟦β⟧(x, y) =>
    match x.step?, y.step? with
    | .some ⟨x', ⟨h₁⟩⟩, _ => .some ⟨(DCC ⟦β⟧(x', y)), ⟨.sigl h₁⟩⟩
    | _, .some ⟨y', ⟨h₁⟩⟩ => .some ⟨(DCC ⟦β⟧(x, y')), ⟨.sigr h₁⟩⟩
    | _, _ => .none
  | DCC (f x) =>
    match f.step?, x.step? with
    | .some ⟨f', ⟨h₁⟩⟩, _ => .some ⟨(DCC (f' x)), ⟨.congrFun h₁⟩⟩
    | _, .some ⟨x', ⟨h₁⟩⟩ => .some ⟨(DCC (f x')), ⟨.congrArg h₁⟩⟩
    | _, _ => .none
  | _ => .none
```

Note that none of our $\rightarrowtail$ rules duplicate their term arguments. Furthermore, each rule (except for congruence) eliminates a combinator. We thus define a syntactic notion of term size:

```
inductive Size : Expr → ℕ → Prop
  | tintro : Size (DCC *) 0
  | top    : Size (DCC ⊤) 0
  | bot    : Size (DCC ⊥) 0
  | subst  : Size x m → Size y n → Size (DCC Subst(x, y)) (m + n).succ
  | sig    : Size (DCC Σ(β)) 0
  | coup   : Size x m → Size y n → Size (DCC ⟦β⟧(x, y)) (m + n)
  | sort   : Size (DCC Sort m) 0
  | pi     : Size (DCC Π(α, β)) 0
  | quot   : Size x m → Size (DCC `(β, x)) m.succ
  | id     : Size (DCC Id(α)) 1
  | fst    : Size (DCC Fst(_α)) 1
  | snd'   : Size (DCC Snd'(δ, β)) 1
  | snd    : Size (DCC Snd(_α)) 1
  | appl   : Size (DCC App(_α, β)) 1
  | comp   : Size f m → Size g n → Size (DCC (f ∘ g)) (m + n).succ
  | pair   : Size f m → Size g n → Size (DCC ⟨f, g⟩) (m + n).succ
  | curr   : Size x m → Size (DCC Λ(α, β, x)) m.succ
  | app    : Size f m → Size x n → Size (DCC (f x)) (m + n)
```

Except for objects with no reduction rules, sizes are given by the successor of the sum of the constituent term point sizes. For example, type points like α in Snd' have no weight, since they are computationally irrelevant.

We compute term size like such:

```
@[aesop norm (rule_sets := [DccSize]), dcc_size, grind]
def size (e : Expr) : ((n : ℕ) × (PULift.{1, 0} (Size e n))) :=
  match e with
  | DCC ⋆ => ⟨_, ⟨.tintro⟩⟩
  | DCC ⊤ => ⟨_, ⟨.top⟩⟩
  | DCC ⊥ => ⟨_, ⟨.bot⟩⟩
  | DCC Subst(x, y) =>
    let ⟨_, ⟨h₁⟩⟩ := x.size
    let ⟨_, ⟨h₂⟩⟩ := y.size
    ⟨_, ⟨.subst h₁ h₂⟩⟩
  | DCC Sort _m => ⟨_, ⟨.sort⟩⟩
  | DCC Σ(_β) => ⟨_, ⟨.sig⟩⟩
  | DCC `(_β, x) =>
    let ⟨_, ⟨h₁⟩⟩ := x.size
    ⟨_, ⟨.quot h₁⟩⟩
  | DCC ⟦_β⟧(x, y) =>
    let ⟨_, ⟨h₁⟩⟩ := x.size
    let ⟨_, ⟨h₂⟩⟩ := y.size
    ⟨_, ⟨.coup h₁ h₂⟩⟩
  | DCC Id(_α)       => ⟨_, ⟨.id⟩⟩
  | DCC Fst(_α) => ⟨_, ⟨.fst⟩⟩
  | DCC Snd'(_α,_β) => ⟨_, ⟨.snd'⟩⟩
  | DCC Snd(_α) => ⟨_, ⟨.snd⟩⟩
  | DCC Π(_α, _β) => ⟨_, ⟨.pi⟩⟩
  | DCC App(_α, _β) => ⟨_, ⟨.appl⟩⟩
  | DCC (f ∘ g) =>
    let ⟨_m, ⟨h₁⟩⟩ := f.size
    let ⟨_n, ⟨h₂⟩⟩ := g.size
    ⟨_, ⟨.comp h₁ h₂⟩⟩
  | DCC ⟨a, b⟩ =>
    let ⟨_o, ⟨h₁⟩⟩ := a.size
    let ⟨_p, ⟨h₂⟩⟩ := b.size
    ⟨_, ⟨.pair h₁ h₂⟩⟩
  | DCC Λ(_α, _β, x) =>
    let ⟨_n, ⟨h₁⟩⟩ := x.size
    ⟨_, ⟨.curr h₁⟩⟩
  | DCC (f x) =>
    let ⟨_, ⟨h₁⟩⟩ := f.size
    let ⟨_, ⟨h₂⟩⟩ := x.size
    ⟨_, ⟨.app h₁ h₂⟩⟩

@[dcc_rel, aesop norm (rule_sets := [DccRel])]
instance Size.instMeasure : Measure Size where
  run := size
```

4.3 Strong Normalization of Affine Reduction

Using our size measure, typeclasses, and tactic automation, we show strong normalization.

Theorem 4. *If e is an expression, then e is strongly-normalizing under $\rightarrowtail$.*

Proof.

```
instance RedAff.instRedRel : RedRel RedAff Size where
  run := step?

theorem red_aff_sn : ∀ (e : Expr), Acc (fun b a => RedAff a b) e :=
  RedAff.instRedRel.sn
```

□

Theorem 5. *If e is an expression, then e is strongly-normalizing under $\rightarrowtail^+$.*

Proof.

```
theorem red_aff_trans_sn : ∀ (e : Expr), Acc
  (fun b a => Relation.TransGen RedAff a b) e :=
  RedAff.instRedRel.to_trans.sn
```

□

4.4 Confluence of Affine Reduction

We demonstrate local confluence by induction on the reduction relation.

Theorem 6. *If $A \rightarrowtail B$ and $A \rightarrowtail C$, then there exists D, such that $B \rightarrowtail^* D$ and $C \rightarrowtail^* D$.*

Proof. We proceed by induction on the reduction relation. We have omitted several repetitive cases for brevity (see (**??**)).

```
open Congruent in @[dcc_reduce, aesop safe apply (rule_sets := [DccReduce])] theorem
↪  RedAff.conf (h₁ : RedAff A B) (h₂ : RedAff A C) : ∃ D, RedAff.instRedRel.ReflTrans B D ∧
↪  RedAff.instRedRel.ReflTrans C D := by
  induction h₁ generalizing C with
  | subst₁p =>
    cases h₂ with
    | subst₁p => exact ⟨_, .refl, .refl⟩
    | congrFun h => cases h with
      | congrFun h => cases h
      | congrArg h => exact ⟨_, congrFun' (congrArg' (.single h)), .single .subst₁p⟩
    | congrArg h => cases h with
      | congrFun h => cases h
      | congrArg h => exact ⟨_, congrArg' (congrArg' (.single h)), .single .subst₁p⟩
  | subst₁ =>
    cases h₂ with
    | subst₁ => exact ⟨_, .refl, .refl⟩
    | congrFun h => cases h with
      | congrFun h => cases h
      | congrArg h => exact ⟨_, congrFun' (congrArg' (.single h)), .single .subst₁⟩
    | congrArg h => cases h with
      | sigl h => exact ⟨_, congrArg' (.single h), .single .subst₁⟩
      | sigr h => exact ⟨_, .refl, .single .subst₁⟩
  | appl =>
    cases h₂ with
    | appl => exact ⟨_, .refl, .refl⟩
    | congrFun h => cases h with
      | congrFun h => cases h
      | congrArg h => exact ⟨_, .refl, .single .appl⟩
    | congrArg h =>
      cases h with
      | sigl h => exact ⟨_, .single (.congrFun h), .single .appl⟩
      | sigr h => exact ⟨_, .single (.congrArg h), .single .appl⟩
  | congrFun h_r ih =>
    cases h₂
    case subst₁p =>
      cases h_r with
      | congrFun h => cases h
      | congrArg h => exact ⟨_, .single .subst₁p, congrFun' (congrArg' (.single h))⟩
    case snd' | quot | id | comp => cases h_r
    case congrFun f f' x f'' h =>
      have ⟨_, h₁, h₂⟩ := ih h
      exact ⟨_, congrFun' h₁, congrFun' h₂⟩
    case congrArg f f' x x'' h_x =>
      exact ⟨_, congrArg' (.single h_x), congrFun' (.single h_r)⟩
    case _ => sorry -- omitted for brevity
  | _ => sorry -- omitted for brevity

@[dcc_rel, aesop norm (rule_sets := [DccRel])] instance RedAff.instConf : Confluent RedAff :=
↪  ⟨RedAff.conf⟩
```

□

Theorem 7. *If $A \rightarrowtail^* B$ and $A \rightarrowtail^* C$, then there exists D, such that $B \rightarrowtail^* D$ and $C \rightarrowtail^* D$.*

Proof. By Newmann's lemma.

```
theorem red_aff_globally_conf : Relation.ReflTransGen RedAff A B
  → Relation.ReflTransGen RedAff A C
  → ∃ D, Relation.ReflTransGen RedAff B D
  ∧ Relation.ReflTransGen RedAff C D :=
  RedAff.instRedRel.globally_confluent.conf
```

□

4.5 Completeness of an affine reducer

Finally, we show semi-completeness of `step?` (4.1) and completeness of `reduce` with respect to normal forms under $\rightarrowtail^*$.

Theorem 8. *If $A \rightarrowtail B$, then there exists B' and a proof h that $A \rightarrowtail B'$ such that step?A = some(B', h).*

Proof.

```
def semi_complete {A B : Expr} (hr : A.RedAff B) : ∃ B', ∃ (h : A.RedAff B'), (step? A) =
↪  .some ⟨B', ⟨h⟩⟩ := by
  change ∃ _, ∃ _, (step? _) = .some _
  unfold step?
  split
  repeat exact ⟨_, (RedAff.instRedRel.sound rfl).down, rfl⟩
  case h_14 =>
    cases hr with
    | sigl h_x =>
      have ⟨_, hrx', heq'⟩ := semi_complete h_x
      exact ⟨_, .sigl (by assumption), by rw [heq']⟩
    | sigr h_y =>
      rename_i x y _
      have ⟨_, hry', heq'⟩ := semi_complete h_y
      match x.step? with
      | .none =>
        exact ⟨_, .sigr (by assumption), by rw [heq']⟩
      | .some ⟨_, ⟨h_x'⟩⟩ =>
        exact ⟨_, .sigl (by assumption), by rw [heq']⟩
  case h_15 => sorry -- omitted for brevity
  case h_16 => sorry -- omitted for brevity

@[dcc_rel, aesop norm (rule_sets := [DccRel])]
instance RedAff.instSemicomplete : SemiComplete RedAff := ⟨semi_complete⟩
```

□

Theorem 9. *If h is a proof that $A \rightarrowtail^* B$ and B is a normal form, then reduce$A$ = (B, h).*

Proof.

```
theorem aff_complete (h : Relation.ReflTransGen RedAff A B)
  (hn : IsNorm RedAff B) : reduce RedAff A = ⟨B, ⟨h⟩⟩ :=
  RedAff.instRedRel.instComplete.complete h hn
```

□

4.6 η-Reduction

To simplify type expressions, we employ Curien's η-reduction rules [4, p. 221], and extend them to Π and Σ. We do not have congruence, and as such inherit deterministic reduction. We define $\rightarrow_\eta$ like such:

```
inductive RedEta : Expr → Expr → Prop
  | curr      : RedEta x x' → RedEta (DCC (Λ(α, β, x))) (DCC (Λ(α, β, x')))
  | curr_quot : RedEta (DCC (Λ(α, Id(α), `(Coup(Id(α)), x)) y)) (DCC `(y, x))
  | curr_fst  : RedEta (DCC (Λ(α, β, x ∘ Fstc(β₂)) y)) (DCC `(β, x))
  | pi        : RedEta (DCC Π(α, β)) p' → RedEta (DCC (Π(α, β) env)) (DCC (p' env))
  | pi_quot   : RedEta x x' → RedEta (DCC (Π(α, `(β, x)))) (DCC (Π(α, `(β, x'))))
  | pi'       : RedEta β β' → RedEta (DCC (Π(α, β))) (DCC (Π(α, β')))
  | sig       : RedEta (DCC (Σ(β))) b' → RedEta (DCC (Σ(β) env)) (DCC (b' env))
  | sig'      : RedEta β β' → RedEta (DCC (Σ(β))) (DCC Σ(β'))
  | idl       : RedEta (DCC (Id(α) ∘ f)) f
  | fst       : RedEta (DCC (Fst(_β) _env) ∘ ⟨x, y⟩) x
  | snd       : RedEta (DCC (Snd(_β) _env) ∘ ⟨x, y⟩) y
  | beta      : RedEta
    (DCC ((App(_f_α, _f_β) _env) ∘ ⟨Λ(α, _β, x), y⟩))
    (DCC (x ∘ ⟨Id(α), y⟩))
  | ai        : RedEta (DCC Λ(α, β, (App(f_α, f_β) env))) (DCC Id((Π(f_α, f_β) env)))
  | fsi       : RedEta (DCC ⟨(Fst(_β) env), (Snd(_β₂) env₂)⟩) (DCC Id((Σ(_β) env)))
  | uncur     : RedEta (DCC Λ(α, β,
    (App(_f_α, _f_β) _env) ∘ ⟨f ∘ (Fst(_x) _env₁), (Snd(_x₂) _env₂)⟩)) f
```

We only η-reduce Π types with respect to their codomains, since their domains generally do not perform significant modification to the environment, since it has already been processed externally. We also omit Curien's *Idr* rule, since it does not occur frequently in our types.

We compute $\rightarrow_\eta$ like such:

```
@[dcc_step, aesop norm (rule_sets := [DccStep])]
def step_eta? (e : Expr) : Option ((e' : Expr) × (PULift.{1, 0} (RedEta e e'))) :=
  match e with
  | DCC (Λ(α, _b, `(_c, x)) y) =>
    if heq : DCC (Λ(α, _b, `(_c, x)) y) = (DCC (Λ(α, Id(α), `(Coup(Id(α)), x)) y)) then
      heq ▸ .some ⟨_, ⟨.curr_quot⟩⟩
    else .none
  | DCC (Λ(_α, _β, _x ∘ Fstc(_β₂)) _y) => .some ⟨_, ⟨.curr_fst⟩⟩
  | DCC Σ(f_β) =>
    match f_β.step_eta? with
    | .some ⟨_, ⟨h⟩⟩ => .some ⟨_, ⟨.sig' h⟩⟩
    | _ => .none
  | DCC (Σ(f_β) _env) =>
    match (DCC Σ(f_β)).step_eta? with
    | .some ⟨_, ⟨h⟩⟩ => .some ⟨_, ⟨.sig h⟩⟩
    | _ => .none
  | DCC (Π(_f_α, f_β) _env) =>
    match (DCC Π(_f_α, f_β)).step_eta? with
    | .some ⟨_, ⟨h⟩⟩ => .some ⟨_, ⟨.pi h⟩⟩
    | _ => .none
  | DCC (Π(_f_α, f_β)) =>
    match f_β with
    | DCC `(_β, x) =>
      match x.step_eta? with
      | .some ⟨_, ⟨h⟩⟩ => .some ⟨_, ⟨.pi_quot h⟩⟩
      | _ => .none
    | x =>
      match x.step_eta? with
      | .some ⟨_, ⟨h⟩⟩ => .some ⟨_, ⟨.pi' h⟩⟩
      | _ => .none
  | (DCC (Id(_α) ∘ _f)) => .some ⟨_, ⟨.idl⟩⟩
  | (DCC (Fst(_β) _env) ∘ ⟨_x, _y⟩) => .some ⟨_, ⟨.fst⟩⟩
  | (DCC (Snd(_β) _env) ∘ ⟨_x, _y⟩) => .some ⟨_, ⟨.snd⟩⟩
  | DCC Λ(α, β, x) =>
    match x.step_eta? with
    | .some ⟨_, ⟨h⟩⟩ => .some ⟨_, ⟨.curr h⟩⟩
    | .none =>
      match DCC Λ(α, β, x) with
      | (DCC Λ(_α, _β, (App(_f_α, _f_β) _env))) => .some ⟨_, ⟨.ai⟩⟩
      | (DCC Λ(_α, _β,
        (App(_f_α, _f_β) _env) ∘ ⟨_f ∘ (Fst(_x) _env₁), (Snd(_x₂) _env₂)⟩)) =>
        .some ⟨_, ⟨.uncur⟩⟩
      | _ => .none
  | (DCC ((App(_f_α, _f_β) _env) ∘ ⟨Λ(__α, __β, _x), _y⟩)) => .some ⟨_, ⟨.beta⟩⟩
  | (DCC ⟨(Fst(_β) _env), (Snd(_β₂) _env₂)⟩) => .some ⟨_, ⟨.fsi⟩⟩
  | _ => .none
```

4.6.1 Syntactic size under η-reduction

We define a size measure for $\to_\eta$, since η-reduction modifies Π codomains, whereas affine reduction does not. The size measure is exactly the same, except type terms are given weights:

```
inductive EtaSize : Expr → ℕ → Prop
  | tintro : EtaSize (DCC *) 1
  | top    : EtaSize (DCC ⊤) 1
  | bot    : EtaSize (DCC ⊥) 1
  | subst  : EtaSize x m → EtaSize y n → EtaSize (DCC Subst(x, y)) (m + n).succ.succ
  | sig    : EtaSize β m → EtaSize (DCC Σ(β)) m.succ
  | coup   : EtaSize x m → EtaSize y n → EtaSize (DCC ⟦β⟧(x, y)) (m + n).succ
  | sort   : EtaSize (DCC Sort m) 1
  | pi     : EtaSize β m → EtaSize (DCC Π(α, β)) m.succ
  | quot   : EtaSize x m → EtaSize (DCC `(β, x)) m.succ.succ
  | id     : EtaSize (DCC Id(α)) 2
  | fst    : EtaSize (DCC Fst(_α)) 2
  | snd'   : EtaSize (DCC Snd'(δ, β)) 2
  | snd    : EtaSize (DCC Snd(_α)) 2
  | appl   : EtaSize (DCC App(_α, β)) 2
  | comp   : EtaSize f m → EtaSize g n → EtaSize (DCC (f ∘ g)) (m + n).succ.succ
  | pair   : EtaSize f m → EtaSize g n → EtaSize (DCC ⟨f, g⟩) (m + n).succ.succ
  | curr   : EtaSize x m → EtaSize (DCC Λ(α, β, x)) m.succ.succ
  | app    : EtaSize f m → EtaSize x n → EtaSize (DCC (f x)) (m + n)
```

We compute η-term size like such:

```
@[aesop norm (rule_sets := [DccSize]), dcc_size, grind]
def eta_size (e : Expr) : ((n : ℕ) × (PULift.{1, 0} (EtaSize e n))) :=
  match e with
  | DCC ⋆ => ⟨_, ⟨.tintro⟩⟩
  | DCC ⊤ => ⟨_, ⟨.top⟩⟩
  | DCC ⊥ => ⟨_, ⟨.bot⟩⟩
  | DCC Subst(x, y) =>
    let ⟨_, ⟨h₁⟩⟩ := x.eta_size
    let ⟨_, ⟨h₂⟩⟩ := y.eta_size
    ⟨_, ⟨.subst h₁ h₂⟩⟩
  | DCC Sort _m => ⟨_, ⟨.sort⟩⟩
  | DCC Σ(β) =>
    let ⟨_, ⟨h₁⟩⟩ := β.eta_size
    ⟨_, ⟨.sig h₁⟩⟩
  | DCC `(_β, x) =>
    let ⟨_, ⟨h₁⟩⟩ := x.eta_size
    ⟨_, ⟨.quot h₁⟩⟩
  | DCC ⟦_β⟧(x, y) =>
    let ⟨_, ⟨h₁⟩⟩ := x.eta_size
    let ⟨_, ⟨h₂⟩⟩ := y.eta_size
    ⟨_, ⟨.coup h₁ h₂⟩⟩
  | DCC Id(_α)      => ⟨_, ⟨.id⟩⟩
  | DCC Fst(_α) => ⟨_, ⟨.fst⟩⟩
  | DCC Snd'(_α,_β) => ⟨_, ⟨.snd'⟩⟩
  | DCC Snd(_α) => ⟨_, ⟨.snd⟩⟩
  | DCC Π(_α, β) =>
    let ⟨_, ⟨h₁⟩⟩ := β.eta_size
    ⟨_, ⟨.pi h₁⟩⟩
  | DCC App(_α, _β)  => ⟨_, ⟨.appl⟩⟩
  | DCC (f ∘ g) =>
    let ⟨_m, ⟨h₁⟩⟩ := f.eta_size
    let ⟨_n, ⟨h₂⟩⟩ := g.eta_size
    ⟨_, ⟨.comp h₁ h₂⟩⟩
  | DCC ⟨a, b⟩ =>
    let ⟨_o, ⟨h₁⟩⟩ := a.eta_size
    let ⟨_p, ⟨h₂⟩⟩ := b.eta_size
    ⟨_, ⟨.pair h₁ h₂⟩⟩
  | DCC Λ(_α, _β, x) =>
    let ⟨_n, ⟨h₁⟩⟩ := x.eta_size
    ⟨_, ⟨.curr h₁⟩⟩
  | DCC (f x) =>
    let ⟨_, ⟨h₁⟩⟩ := f.eta_size
    let ⟨_, ⟨h₂⟩⟩ := x.eta_size
    ⟨_, ⟨.app h₁ h₂⟩⟩

@[dcc_rel, aesop norm (rule_sets := [DccRel])]
instance EtaSize.instMeasure : Measure EtaSize where
  run := eta_size
```

4.6.2 η Completeness

We derive strong normalization from a `RedRel` instance.

Theorem 10. *If e is an expression, then e is strongly normalizing under $\to_\eta$.*

Proof.

```
@[dcc_rel, aesop norm (rule_sets := [DccRel])]
instance RedEta.instRedRel : RedRel RedEta EtaSize where
  run := step_eta?

theorem red_eta_sn : ∀ (e : Expr), Acc (fun b a => RedEta a b) e := RedEta.instRedRel.sn
```

□

We prove confluence by demonstrating that single-step η-reduction is deterministic.

Theorem 11. *If $A \to_\eta B$ and $A \to_\eta C$, then $B = C$.*

Proof. We proceed by induction on the reduction relation.

```
@[simp] theorem RedEta.unique (h₁ : RedEta A B) (h₂ : RedEta A C) : B = C := by
  induction h₁ generalizing C with
  | sig' | pi | curr_fst | sig => cases h₂; aesop
  | pi_quot hr ih =>
```

```
    cases h₂
    · rename_i hr₂; aesop
    · rename_i hr₂; cases hr₂
  | pi' hr ih =>
    cases h₂
    · rename_i hr₂; cases hr
    · aesop
  | curr hr ih =>
    cases h₂ with
    | curr hr₂ => rw [ih hr₂]
    | ai | uncur => cases hr
  | ai | uncur =>
    cases h₂
    · rename_i h; cases h
    · rfl
  | curr_quot | snd | fst | idl | beta | fsi => cases h₂; rfl
```

□

Theorem 12. *If $A \to_\eta B$ and $A \to_\eta C$, then there exists D, such that $B \to_\eta^* D$ and $C \to_\eta^* D$.*

Proof. (4.6.2) By uniqueness under $\to_\eta$:

```
theorem RedEta.conf (h₁ : RedEta A B) (h₂ : RedEta A C) :
  ∃ D, Relation.ReflTransGen B D
  ∧ Relation.ReflTransGen C D := by
  have h := h₁.unique h₂
  exact ⟨B, h ▸ .refl, h ▸ .refl⟩
```

□

Theorem 13. *If $A \to_\eta^* B$ and $A \to_\eta^* C$, then there exists D, such that $B \to_\eta^* D$ and $C \to_\eta^* D$.*

Proof. By Newmann's lemma.

```
@[dcc_rel, aesop norm (rule_sets := [DccRel])]
instance RedEta.instConf : Confluent RedEta := ⟨RedEta.conf⟩

theorem red_eta_globally_conf : Relation.ReflTransGen RedEta A B → Relation.ReflTransGen
↪  RedEta A C → ∃ D,
  Relation.ReflTransGen RedEta B D ∧ Relation.ReflTransGen RedEta C D :=
  RedEta.instRedRel.globally_confluent.conf
```

□

Due to determinism, we have completeness under single-step $\to_\eta$.

Theorem 14. *If h is a proof that $A \to_\eta B$, then $step_eta?A = some(B, h)$.*

Proof. We proceed by induction on the step relation.

```
theorem RedEta.complete (h_red : RedEta A B) : step_eta? A = .some ⟨B, ⟨h_red⟩⟩ := by
  induction h_red <;> (
    unfold step_eta?
    aesop (rule_sets := [DccStep, DccReduce, DccSize, DccRel]))
```

□

As in $\rightarrowtail$, we inherit completeness of full reduction with respect to $\to_\eta^*$ and normal forms.

Theorem 15. *If h is a proof that $A \to_\eta^* B$ and B is a normal form, then $reduceA = (B, h)$.*

Proof. By typeclass automation:

```
def eta_semi_complete {A B : Expr} (h_red : A.RedEta B) :
  ∃ B', ∃ (h : A.RedEta B'), (step_eta? A) = .some ⟨B', ⟨h⟩⟩ :=
  ⟨_, _, h_red.complete⟩

@[dcc_rel, aesop norm (rule_sets := [DccRel])] instance RedEta.instSemicomplete : SemiComplete
↪  RedEta := ⟨eta_semi_complete⟩

theorem eta_complete (h : Relation.ReflTransGen RedEta A B)
  (hn : IsNorm RedEta B) : reduce RedEta A = ⟨B, ⟨h⟩⟩ :=
  RedEta.instRedRel.instComplete.complete h hn
```

□

5 Type Inference

Note that the types of all of the dependent categorical combinators are affine (3.3). Hence, we use our decision procedure (**??**) for $=_\beta$ under affine-reduction to prove decidability of type-checking.

We do not normalize inside our judgment relation, nor do we have a conversion case. We bake conversion into each applicable constructor with a `Reduces` predicate by normalizing the expected and actual types a la carte:

```
inductive Reduces : Expr → Expr → Type
  | mk : RedAffReflTrans A B → IsNormAff B → Reduces A B
deriving Repr

theorem Reduces.complete : Reduces A B → reduce A = B :=
  fun ⟨ha, hb⟩ => ha.complete hb

def reduce_reduces (h : reduce A = B) : Reduces A B := ⟨reduce_sound h, reduce_norm h⟩

def reduce_reduces' : Reduces A A.reduce := reduce_reduces rfl
```

For example, application is inferred like such:

```
inductive Judge : Expr → Expr → Type where
  | app   : Judge f t_f
    → Reduces t_f (DCC (Π(m, n) α β))
    → Judge x t_x
    → Reduces t_x α
    → Judge (DCC (f x)) (DCC (β ⟦Sort m, Id(Sort m)⟧(α, x)))
```

We defer full type normalization to a supplementary predicate, `JudgeConv`:

```
inductive JudgeConv : Expr → Expr → Type
  | mk : Judge x A → Reduces A B → JudgeConv x B
```

We define an `IsArrow` predicate for convenience in pattern matching arrows, which are by convention, Π-objects containing quotation in their codomain. We do not assume Π-extensionality.

```
inductive IsArrow : Expr → (α β : Expr) → Level → Level → Type
  | mk    : Reduces A (arrow α β m n) → IsArrow A α β m n
deriving Repr

def is_arrow (t : Expr) : Option ((α β : Expr) × (m n : Level) × IsArrow t α β m n) :=
  match heq₀ : t.reduce with
  | DCC (Π(m, n) α `(Sort n₂, Σ(Sort m₂, Id(Sort m₃)), β)) =>
    if heq : m = m₂ ∧ m = m₃ ∧ n = n₂ then by
      have ⟨ha, hb, hc⟩ := heq
      rw [← ha, ← hb, ← hc] at heq₀
      refine .some ⟨α, β, m, n, ?_, ?_⟩
      · exact reduce_sound heq₀
      · exact reduce_norm heq₀
    else .none
  | _ => .none
```

Our `Judge` relation models the inference rules (3.3). We have omitted the individual types and corresponding constructors for brevity, but give the constructor for tail-dependent Curry (**??**) and App (17) as examples. See the appendix and supplementary materials for the full formalization.

```
def curr.type (α γ : Expr) (m n : Level) : Expr :=
  (DCC (Π(m.succ, n) α ((Flip(Π(m, n)) (γ ∘'
  (Λ'(Sort m.succ, Id(Sort m.succ), Id(Σ(Sort m.succ, Id(Sort m.succ))))
    ⟦Sort m.succ, Id(Sort m.succ)⟧(α, Id(α))))) ∘' Snd'(Sort m.succ, α))))

def app_e.type (α β : Expr) (m n : Level) : Expr :=
  let t_f := DCC (Π(m, n) α β)
  let t_x := DCC `(Sort n, t_f, α)

  DCC (Π(max m n, n)
      Σ(t_f, t_x)
      ((Λ'(Sort m, Id(Sort m), β) α) ∘' (Snd(t_f, t_x) ∘ Snd'(Sort (max m n), Σ(t_f, t_x)))))

inductive Judge : Expr → Expr → Type where
  | curr  : Judge α t_α
    → Reduces t_α (DCC (Sort m.succ))
```

```
    → Judge x t_x
    → Reduces t_x (DCC (Π(m.succ, n) Σ(α, Id(α)) γ))
    → Judge (DCC λ(α, x)) (curr.type α γ m n)
  | app_e : Judge α t_α
    → Reduces t_α (DCC Sort m)
    → Judge (DCC App(n, α, β)) (app_e.type α β m n)
  | app   : Judge f t_f
    → Reduces t_f (DCC (Π(m, n) α β))
    → Judge x t_x
    → Reduces t_x α
    → Judge (DCC (f x)) (DCC (β ⟦Sort m, Id(Sort m)⟧(α, x)))
  | sort  : Judge (DCC Sort m) (DCC Sort m.succ)
deriving Repr
```

5.1 Computing Type Inference

Using our terminating affine β-reduction function and `reduce_reduces`, we define an `infer` function with soundness. We give only the cases described in our `Judge` illustration, though the rest are available in supplementary materials. Pattern matching against our `Judge` constructors is somewhat obtuse, primarily due to our design decision to avoid adding a conversion constructor and reducing the inferred types.

```
def infer (e : Expr) : Option ((t : Expr) × (Judge e t)) :=
  match e with
  | DCC Sort m => .some ⟨DCC Sort m.succ, .sort⟩
  | DCC App(n, α, β) =>
    match α.infer with
    | .some ⟨t_α, h_t_α⟩ =>
      match heq₀ : t_α.reduce with
      | DCC Sort m =>
        .some ⟨_, .app_e h_t_α (reduce_reduces heq₀)⟩
      | _ => .none
    | _ => .none
  | DCC λ(α, x) =>
    match heq₀ : α.infer, x.infer with
    | .some ⟨t_α, h_t_α⟩, .some ⟨t_x, h_t_x⟩ =>
      match heq₁ : t_α.reduce with
      | DCC Sort m + 1 =>
        match heq₃ : t_x.reduce with
        | (DCC (Π(m₂ + 1, n) Σ(α₁, Id(α₂)) γ₂)) =>
          if heq₂ : m₂ = m ∧ α₁ = α ∧ α = α₂ then
            have ⟨ha, hb, hc⟩ := heq₂
            .some ⟨_, Judge.curr
              h_t_α
              (reduce_reduces heq₁)
              h_t_x
              (ha ▸ hb ▸ hc ▸ (reduce_reduces heq₃))⟩
          else .none
        | _ => .none
      | _ => .none
    | _, _ => .none
  | DCC (f x) => match f.infer, x.infer with
    | .some ⟨t_f, h_f⟩, .some ⟨t_x, h_x⟩ =>
      match heq₀ : t_f.reduce, heq₁ : t_x.reduce with
      | DCC (Π(m, n) α₁ β), α₂ =>
        if heq : α₁ = α₂ then
          .some ⟨(DCC (β ⟦Sort m, Id(Sort m)⟧(α₁, x))),
            .app h_f (reduce_reduces heq₀)
                 h_x (reduce_reduces (heq ▸ heq₁))⟩
        else .none
      | _, _ => .none
    | _, _ => .none
  | _ => sorry -- remaining cases omitted for brevity

def infer_conv (e : Expr) : Option ((t : Expr) × JudgeConv e t) :=
  match infer e with
  | .some ⟨t, h⟩ =>
    .some ⟨t.reduce, ⟨h, reduce_reduces rfl⟩⟩
  | _ => .none
```

Avoiding a conversion constructor proves to be quite advantageous, since neither confluence or induction are necessary, and case analysis suffices to prove useful properties. For example, a proof of preservation of $Id(\alpha)x$ is quite straightforward.

Theorem 16. *For all $\alpha \in Expr$ and $x \in Expr$, if $Id(\alpha)\ x : reduce(\alpha)$, then $x : reduce(\alpha)$.*

Proof. We proceed by case analysis on the judgment $Id(\alpha)\ x : reduce(\alpha)$.

```
def id_preservation (h : JudgeConv (DCC (Id(α) x)) α.reduce) : (JudgeConv x α.reduce) :=
  match h with
  | ⟨.app (.id _ _) b h_t_x _, h_red⟩ => by
    cases reduce_arrow b.complete
    exact ⟨h_t_x, by assumption⟩
```

□

5.1.1 Infer Completeness

Yet another advantage of avoiding a conversion constructor, and of having decidable $\beta_=$ is that completeness is provable. We have omitted all but the `curr`, `app_e`, `sort`, and `app` constructors (see supplementary materials).

Theorem 17. *For all $e \in Expr$ and $t \in Expr$, if $e : t$, then `infer e = some t`.*

Proof. We proceed by parallel induction on the typing derivation and case analysis on the unfolding of `infer`.

```
theorem Judge.complete (h_t : Judge e t) : infer e = .some ⟨t, h_t⟩ := by
  induction h_t with
  | sort =>
    unfold infer
    rfl
  | curr h_t_α h_red₁ h_t_x h_red₂ ih₁ ih₂ =>
    have ha := h_red₁.complete
    have hb := h_red₂.complete
    unfold infer
    rw [ih₁, ih₂]
    simp
    split
    case h_1 heq =>
      rw [heq] at ha
      cases ha
      split
      case h_1 heq =>
        rw [heq] at hb
        cases hb
        simp
        split
        rfl
      case h_2 h =>
        exact (h _ _ _ _ _ hb).elim
    case h_2 h =>
      exact (h _ ha).elim
  | app_e h_t_α h_red₁ ih =>
    have ha := h_red₁.complete
    unfold infer
    rw [ih]
    simp
    split
    case h_1 heq =>
      rw [heq] at ha
      cases ha
      rfl
    case h_2 h =>
      exact (h _ ha).elim
  | app h_t_f h_red_f h_t_x h_red_x ih₁ ih₂ =>
    have ha := h_red_f.complete
    have hb := h_red_x.complete
    unfold infer
    simp [ih₁, ih₂]
    split
    case h_1 heq =>
      rw [heq] at ha
      cases ha
      cases hb
      simp
      aesop
    case h_2 h =>
      exact (h _ _ _ _ ha).elim
```

□

5.2 Judgmental Decidability

Using our proof of completeness and soundness proof produced by `infer`, we demonstrate judgmental decidability.

Theorem 18. *For all $e \in Expr$ and $t \in Expr$, $e : t \vee \neg e : t$.*

Proof.

```
def Judge.decide : Judge e t ⊕ (Judge e t → PLift False) :=
  if h : (infer e).isSome then
    if heq : ((infer e).get h).fst = t then
      .inl (heq ▸ ((infer e).get h).snd)
    else
      .inr fun hn => by
        have h_comp := hn.complete
        simp_all
  else .inr fun hn => by simp [hn.complete] at *

def JudgeConv.decide : JudgeConv e t ⊕ (JudgeConv e t → PLift False) :=
  if h : (infer e).isSome then
    if heq : ((infer e).get h).fst.reduce = t then
      .inl ⟨((infer e).get h).snd, reduce_reduces heq⟩
    else .inr fun ⟨ha, hb⟩ => by
      have h := ha.complete
      simp_all
      exact (heq hb.complete).elim
  else .inr fun ⟨ha, hb⟩ => by simp [ha.complete] at *
```

□

6 Future Work

We hope to use our inference and reduction machinery to mechanize a proof of strong normalization of the entire calculus in the future. That is, a proof of strong normalization for the calculus including the pair combinator, which is not affine.

7 Conclusions

Our formalization gives a novel extension of the simply-typed categorical combinators [4, p. 220] to a dependently-typed setting. By mechanizing the theory in Lean 4, we provide several indispensable guarantees about the logic. Most importantly, we demonstrate a mechanism for deciding type inference of the calculus. We demonstrate strong normalization of an affine restriction on the categorical combinators, and use this property to prove soundness and completeness of an inference function with respect to typing judgments. In doing so, we also demonstrate a useful technique to avoid conversion in the typing relation, given we have decidable $\beta_{\to *}$.

Our proof of judgmental decidability is made possible in principle by the observation that the types of the dependent categorical combinators are purely affine. This fact is not true of equivalent systems, like the dependent SK combinators.

Finally, since the categorical combinators are well-known to be a performant compilation target, we believe that our verified implementation could have ramifications towards the efficient execution of dependent type inference in a purely contextless way.

Acknowledgements

To Jake, for teaching me lambda-calculus.

A Supplementary Materials

Lean 4 formalization repository: https://github.com/lexzaiello/dcc
Omitted inference cases: https://github.com/lexzaiello/DCC/blob/master/Dcc/CC/Judge.lean
Omitted reduction cases: https://github.com/lexzaiello/DCC/blob/master/Dcc/CC/Step2.lean

B Todos left

- Notation inconsistencies: I mix and match $f(x, y)$ and fxy. fxy is the preferred notation. Mixing and matching notation for our calculus and normal inference rule notation is also ambiguous and confusing. How to handle this?
- My notation for application is ugly. Usually application is written as fx, not $f\ x$
- More background: want a reference for telescope
- More background: want a reference for Tarski universes that isn't ncatlab
- If time permits, I want to finish the λ-calculus compiler.

References

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